

PALLADIUM BANK DATA WAREHOUSE PROJECT

1. DESIGN STRATEGY (THE STAR SCHEMA)

The core of this solution is a star schema dimensional report. Unlike a normalized database that is used for daily banking operations, this design will help query performance and give simplicity for users who venture into business.

- **Fact table (fact transactions);** This is the center of the model capturing every individual transaction. Its grain is defined at the transaction level (txn_id), ensuring that no data is loss. By storing numeric measures like amount, fee_amount alongside foreign_keys, it allows for a lot of aggregation across millions of rows.
- **Dimension table:** surrounding the fact table, there are five descriptive dimensions: Dim_Customer, Dim_Product, Dim_Branch, Dim_Channel and Dim_Date. This allows bank to slice into every revenue by using any of the attributes such as the tier, product type, channel_name without going complex.

2. HANDLING DATA THROUGH SCD (Slowly Changing Dimensions)

By adding effective date, end date and is current column in the customers table, the model shows a snapshot of every customer to see whether they're active or not. When a customer upgrades the tier, a new record is created rather than deleting the old record, with this if the head of retail banking request for revenue tier for 2024 the report, the report correctly takes the attribute fees to the customer tier at that specific time and not their current tier as of today.

3. DATA QUALITY AND INTEGRATION

While importing the raw Dataset, we observed a significant formatting issue which was the currency string of “45,000.00” that PostgreSQL couldn't calculate. We use the following;

- **ETL Approach:** we use the Extract, Load and Transform strategy. Data was first loaded into the temporary table(staging_transactions) tables that had flexible varchar types.
- **Transformation:** During the Migration from the staging to the final star schema, we applied logics to remove commas and cast strings into NUMERIC (15, 2) formats. This automated cleaning ensures that the final Fact table was clean and ready for calculation.
- **Quality checks:** To Maintain Integrity, we did the following to established for automated checks;
 - ✓ **Duplicate Detection:** ensures the txn_id is unique.
 - ✓ Validating that every transaction has a matching customer and product key.
 - ✓ **Null Validation:** Flagging any transaction with missing date or amount.
 - ✓ **Value Range:** Checking for negative amount.

4. PERFORMANCE OPTIMIZATION

To support high-speed analytics, we implemented two primary performance layers:

- **Indexing:** We applied indexes on all foreign keys (customer_sk_fk, date_key_fk). This ensures that even if the data has millions of records overtimes, the database can see the correct records in milliseconds rather than scanning the entire table.
- **Aggregation Tables:** We created a Summary table that pre calculates monthly revenue and transaction counts by branch. This data allows for calculation from data that comes in without having to calculate the 18 – month history data every time the report is opened.

5. MEETING THE BUSINESS OBJECTIVES

The model was successfully validated against the bank's core objectives:

- **Churn Signal:** By using the MAX (txn_date) logic in our analytical queries, we can identify high-value customers who haven't done transactions in 30 days providing a recency alert for the churn team.

- **Product Performance:** The model clearly identifies which products drive Deposits vs Withdrawals by joining the Fact table with the Dim_Product, providing a clear view of liquidity and the product popularity.
- **Drill-Downs:** We identified hierarchies (Year-Month-Date and State-Branch) allowing users to have an overview and drill-down into specific data points causing a trend.

6. CONCLUSION

This Schema Architecture we built is more than just collection of tables, it is an analytics platform where SCD could be used for historical accuracy and star schema for speed, staging data for data quality, therefore the Palladium bank has a single version for truth which could be used as guidance for its retail banking strategy for the next 18 months and beyond.